# Math 110 — Week 3

## Solutions

**HW3 P1.**

**Idea.** A formula with dim null plus another dimension is rank–nullity for some map. The left side is $\dim T^{-1}(U)$, so the map should have domain $T^{-1}(U)$: restrict $T$ to it.

1. (a) $T0 = 0 \in U$. If $Tv, Tw \in U$ then $T(\lambda v + w) = \lambda Tv + Tw \in U$.
2. (b) Let $S\colon T^{-1}(U) \to W$, $Sv = Tv$. Its domain is finite-dimensional (a subspace of $V$).
3. $\operatorname{null} S = \operatorname{null} T$: if $Tv = 0$ then $Tv \in U$, so $v \in T^{-1}(U)$; so $\operatorname{null} T \subseteq T^{-1}(U)$ and $\operatorname{null} S = \{v \in T^{-1}(U) : Tv = 0\} = \operatorname{null} T$.
4. $\operatorname{range} S = U \cap \operatorname{range} T$: $Sv = Tv$ lies in $U$ and in $\operatorname{range} T$. Conversely if $w \in U \cap \operatorname{range} T$, then $w = Tv$ with $Tv \in U$, so $v \in T^{-1}(U)$ and $w = Sv$.
5. Rank–nullity for $S$: $\dim T^{-1}(U) = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T)$.
6. (c) The space is $D^{-1}(U)$ with $U = \operatorname{span}(1, x^2)$. $\operatorname{null} D = \operatorname{span}(1)$, dimension 1. $\operatorname{range} D = \mathcal{P}_3(\mathbb{R}) \supseteq U$, so $U \cap \operatorname{range} D = U$, dimension 2. Answer: $1 + 2 = 3$. (A basis: $1,\ x,\ x^3$.)

**HW3 P2.**

**Idea.** $N$ on $\mathbb{R}^2$ has $\operatorname{null} N = \operatorname{range} N$; use two copies. For (b), rank–nullity makes the dimension even.

1. (a) Define $T$ on the standard basis (Prop. 21): $Te_1 = 0$, $Te_2 = 0$, $Te_3 = e_1$, $Te_4 = e_2$, i.e. $T(x_1, x_2, x_3, x_4) = (x_3, x_4, 0, 0)$. Then $\operatorname{range} T = \operatorname{span}(e_1, e_2)$ and $\operatorname{null} T = \{x_3 = x_4 = 0\} = \operatorname{span}(e_1, e_2)$.
2. (b) If $\operatorname{null} T = \operatorname{range} T$, then $5 = \dim \operatorname{null} T + \dim \operatorname{range} T = 2 \dim \operatorname{null} T$, which is impossible since 5 is odd.

**HW3 P3.**

**Idea.** The two conditions are the two coordinates of a linear map to $\mathbb{R}^2$; if it is surjective the null space has dimension $5 - 2$.

1. $S(p) = (p(1), p'(1))$ is linear (evaluation and differentiation are), and $\operatorname{null} S = U$.
2. $S$ is surjective: $S(1) = (1, 0)$ and $S(x - 1) = (0, 1)$ span $\mathbb{R}^2$.
3. $\dim U = \dim \mathcal{P}_4(\mathbb{R}) - \dim \operatorname{range} S = 5 - 2 = 3$.
4. Homework 2 found the basis $(x-1)^2, (x-1)^3, (x-1)^4$ directly. Rank–nullity gives the dimension without finding a basis.

**HW3 P4.**

**Idea.** A linear map sends a relation to a relation with the same coefficients.

1. (a) If $\sum \lambda_j v_j = 0$ with not all $\lambda_j = 0$, then

$$\sum \lambda_j T v_j = T(\sum \lambda_j v_j) = T0 = 0$$

with the same coefficients.

2. (b) $T = 0\colon \mathbb{R} \to \mathbb{R}$ sends the independent list 1 to the dependent list 0. (Or $P(x, y) = (0, y)$ sends $e_1, e_2$ to $0, e_2$.)

### HW3 P5.

**Idea.** Apply $T$ to each monomial; the results are the columns.

1. (a) $T(1) = 1$. $T(x) = x^2 + x$. $T(x^2) = 2x^3 + x^2$. $T(x^3) = 3x^4 + x^3$. So
$$\mathcal{M}(T) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$
2. (b) For $p = a + bx + cx^2 + dx^3$: $Tp = a + bx + (b + c)x^2 + (2c + d)x^3 + 3dx^4$. If $Tp = 0$: $a = 0$, $b = 0$, then $c = 0$, then $d = 0$. So $\operatorname{null} T = \{0\}$, dimension 0.
3. By rank–nullity $\dim \operatorname{range} T = 4$, and $\operatorname{range} T = \operatorname{span}(1,\ x + x^2,\ x^2 + 2x^3,\ x^3 + 3x^4)$.
4. (c) $\operatorname{range} T$ is spanned by the images of the basis vectors $1, x, x^2, x^3$. Column $j$ is the coordinate vector of the $j$-th image. So in coordinates the range is the span of the columns.

### HW3 P6.

**Idea.** Use the second equation to eliminate $z$; in $\mathbb{F}_2$, $x + x = 0$.

1. $x + z = 0$ gives $z = -x = x$.
2. Then $x + y + z + w = 2x + y + w = y + w = 1$, so $w = y + 1$.
3. $x, y$ are free. Solutions $(x, y, z, w)$: $(0, 0, 0, 1)$, $(0, 1, 0, 0)$, $(1, 0, 1, 1)$, $(1, 1, 1, 0)$.

### HW3 P7.

**Idea.** Two solutions differ by an element of the null space, and a $k$-dimensional space over $\mathbb{F}_2$ has $2^k$ elements.

1. Fix one solution $x_0$. Then $Ax = b \iff A(x - x_0) = 0 \iff x \in x_0 + \operatorname{null} A$. So the number of solutions is the number of elements of $\operatorname{null} A$.
2. $A$ is a map $\mathbb{F}_2^m \to \mathbb{F}_2^n$, so $\dim \operatorname{null} A = m - \operatorname{rank} A$.
3. A space of dimension $k$ over $\mathbb{F}_2$ has $2^k$ elements (each of $k$ coordinates is 0 or 1). Answer: $2^{\,m - \operatorname{rank} A}$.

### R1.

**Idea.** Columns are the images of the vectors of $B$, written in $1, x, x^2$. Equal dimensions: injective is enough.

1. $T(1, 0, 0) = 1 + x^2$, $T(1, 1, 0) = 2 + x + x^2$, $T(1, 1, 1) = 2 + 2x + 2x^2$. So
$$\mathcal{M}(T) = \begin{pmatrix} 1 & 2 & 2 \\ 0 & 1 & 2 \\ 1 & 1 & 2 \end{pmatrix}.$$
2. (b) If $T(a, b, c) = 0$: $a + b = 0$, $b + c = 0$, $a + c = 0$. Then $a = -b = c$ and $a + c = 2a = 0$, so $a = b = c = 0$. $T$ is injective.
3. $\dim \mathbb{R}^3 = 3 = \dim \mathcal{P}_2(\mathbb{R})$, so $T$ is also surjective (Prop. 25).

### R2.

**Idea.** Same as HW3 P1(c).

$$\dim D^{-1}(U) = \dim \operatorname{null} D + \dim(U \cap \operatorname{range} D) = 1 + 2 = 3,$$

since $\operatorname{range} D = \mathcal{P}_3(\mathbb{R}) \supseteq U$. Basis: 1 (null space), $x$ ($Dx = 1$), $x^3$ ($Dx^3 = 3x^2$). They have different degrees, so they are independent.

**R3.**

**Idea.** Apply $T$ to $1, x, x^2, x^3$.

$T(1) = 0$, $T(x) = 1$, $T(x^2) = 2 + 2x$, $T(x^3) = 6x + 3x^2$. So

$$\mathcal{M}(T) = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

**R4.**

**Idea.** Count dimensions.

No. That null space has dimension 1, so rank–nullity would give $\dim \operatorname{range} T = 4 - 1 = 3$. But $\operatorname{range} T \subseteq \mathbb{R}^2$ has dimension at most 2.